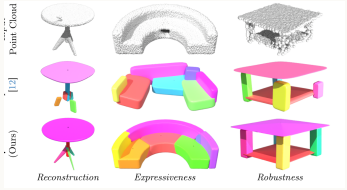
GEOMETRIC DEEP LEARNING

SuperFlex

Deformable Superquadrics for Point Cloud Decomposition

Gabriel Tavernini, Elisabetta Fedele, Tiago Novello, Leonidas Guibas, Marc Pollefeys, Francis Engelmann

ECCV 2026



Tavernini et al., ECCV 2026, Fig. 1.

Contents

1 Motivation and previous work

2 Superquadric geometry

3 Architecture and learning

4 Refinement and partial observations

5 Experimental evaluation

6 Conclusions

Understanding a 3D object through a few geometric parts

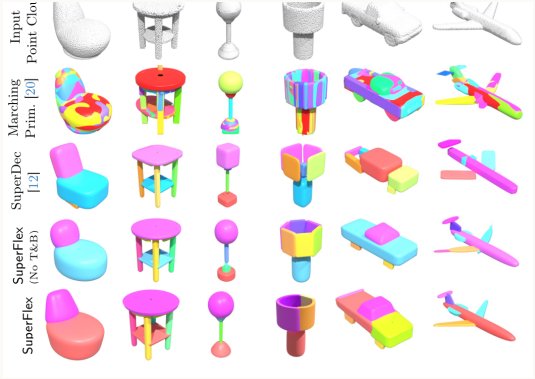
Input

The input is a point cloud $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^{N}$ with $\mathbf{x}_i \in \mathbb{R}^3$.

Output

The output is a compact set of parametric solids whose union approximates the object through meaningful geometric parts.

- **Fidelity** requires a close match to the target geometry.
- **Parsimony** favours a small number of primitives.
- **Robustness** matters when parts of the surface are missing.



Paper Fig. 4. Each colour identifies a distinct primitive.

From earlier methods to SuperFlex

Optimization-based

EMS (2022) introduced probabilistic fitting with improved robustness to outliers

Marching Primitives (2023) uses iterative fitting guided by signed distance fields.

Learning-based

SQ (2019) learned superquadric shape abstractions.

CSA (2021) jointly estimated cuboids and point assignments

SuperDec (2025) introduced Transformer-based set prediction.

SuperFlex (2026)

Inherits SuperDec's point-encoder + Transformer decoder.

Adds expressive geometry, stronger losses, optional refinement and occlusion training.

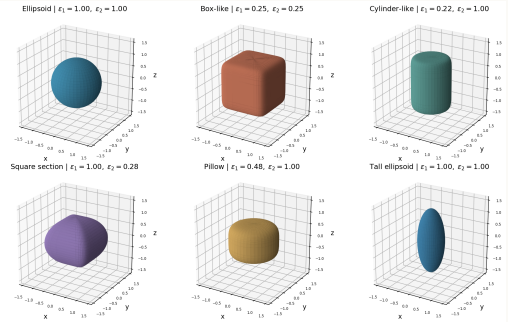
Superellipsoids

Canonical implicit surface

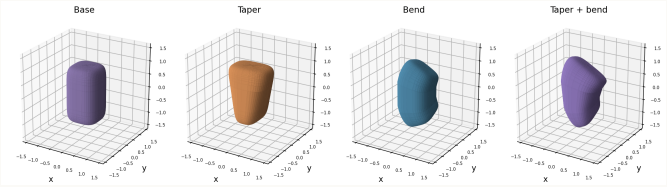
$$q_{\text{can}}(\mathbf{x}) = \left[\left| \frac{x}{s_x} \right|^{\frac{2}{\epsilon_2}} + \left| \frac{y}{s_y} \right|^{\frac{2}{\epsilon_2}} \right]^{\frac{\epsilon_2}{\epsilon_1}} + \left| \frac{z}{s_z} \right|^{\frac{2}{\epsilon_1}}$$

$$\mathcal{S} = \{\mathbf{x} : q_{\text{can}}(\mathbf{x}) = 1\}$$

- $\mathbf{s} = (s_x, s_y, s_z)$ sets the three axis scales.
- ϵ_1 controls the meridional profile.
- ϵ_2 controls the horizontal cross-section.



Five canonical shape parameters are not enough



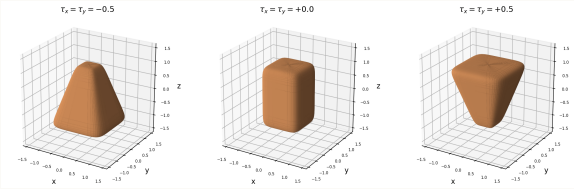
Failure of rigid primitives

Straight, symmetric parts are easy to model. Curved or tapered parts, however, can require several rigid primitives even when they correspond to a single meaningful component.

- **Tapering** changes the cross-sectional width along an axis.
- **Bending** produces a curved central axis.

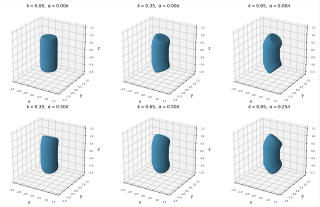
Tapering and bending

Tapering along z



$$x' = x(1 + \tau_x z/s_z),$$
$$y' = y(1 + \tau_y z/s_z), \quad z' = z.$$

Bending about z



- k_z controls curvature, with circular radius $1/|k_z|$.
- α_z sets the orientation of the bending plane.
- Repeat for bending about x and y.

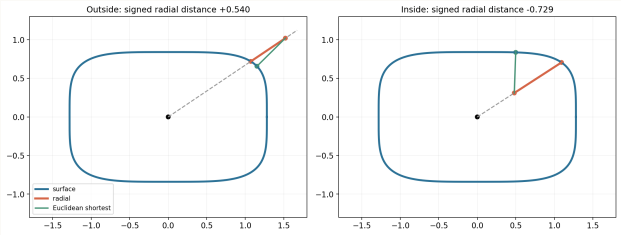
Nineteen parameters, one implicit field

Pose	Scale	Shape	Taper	Bend	**Total**
$(\mathbf{t}, R)$	$\mathbf{s}$	(ϵ_1, ϵ_2)	(τ_x, τ_y)	$\beta = (k_a, \alpha_a)_{a=x,y,z}$	
$3+3$	3	2	2	6	**19**

$$q(\mathbf{x};\Theta) = q_{\text{can}}\left(\underbrace{T^{-1} \circ B_y^{-1} \circ B_x^{-1} \circ B_z^{-1}}_{\text{undo deformations}}(R^\top(\mathbf{x}-\mathbf{t})); \mathbf{s}, \boldsymbol{\epsilon}\right)$$

- Evaluating a world-space point requires undoing its pose and geometric deformations.
- On the surface $q(\mathbf{x};\Theta) = 1$; inside $q < 1$ (in the valid invertible regime).

Radial distance as an efficient geometric proxy



For the rigid case

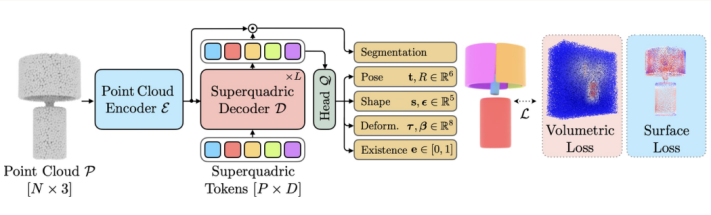
$q(\lambda \mathbf{x}) = \lambda^{2/\epsilon_1} q(\mathbf{x})$

The surface is reached when $q(\lambda_* \mathbf{x}) = 1$, giving
$\lambda_* = q(\mathbf{x})^{-\epsilon_1/2}$

$$\boxed{d_{rad} = \|R^\top(\mathbf{x} - \mathbf{t})\|(1 - q(\mathbf{x}; \Theta)^{-\epsilon_1/2})}$$

- The sign distinguishes exterior points ($d > 0$) from interior points ($d < 0$).
- This efficient proxy is not an exact Euclidean SDF.

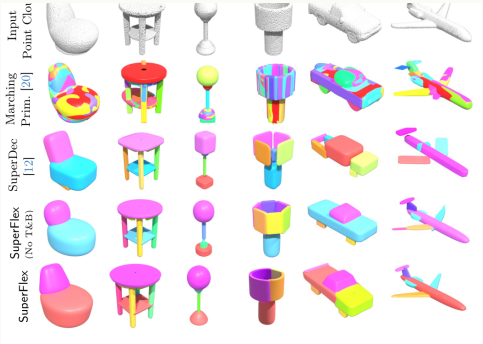
The model architecture



- The input point cloud is $\mathcal{P} \in \mathbb{R}^{N \times 3}$.
- A PVCNN encoder extracts point features $F_{PC} \in \mathbb{R}^{N \times D}$.
- P positional queries represent candidate superquadrics, with $F_{SQ} \in \mathbb{R}^{P \times D}$.
- L Transformer layers combine query self-attention and point-feature cross-attention.
- Prediction heads return $\{(\Theta_p, e_p)\}_{p=1}^{P}$ and soft point assignments.

$$W = \text{softmax}_p(F_{PC}\, \phi(F_{SQ})^\top) \in \mathbb{R}^{N \times P}$$

Why Chamfer distance alone is insufficient



Paper Fig. 4.

SuperDec compared with rigid and deformable SuperFlex.

Chamfer distance

Chamfer distance aligns samples on the observed and predicted surfaces, without directly comparing the volumes of their unions.

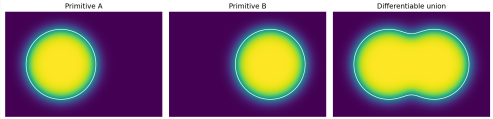
- Shape boundaries can become overly rounded.
- Adjacent primitives can leave gaps.
- Global volume and local surface accuracy are not jointly constrained.

The proposed loss combines global occupancy and local surface alignment.

Global volumetric consistency

1. Convert the radial-distance proxy into **soft occupancy**
$$I_p(\mathbf{x}) = \sigma\left(-\frac{d_p(\mathbf{x})}{T_{\mathrm{IoU}}}\right).$$
2. Combine primitive occupancies with a **probabilistic OR**
$$\hat{O}(\mathbf{x}) = 1 - \prod_{p=1}^{P}(1 - e_p I_p(\mathbf{x})).$$
3. Compare with ground-truth occupancy O over uniform samples $\mathcal{V}$.

$$\mathcal{L}_{\mathrm{IoU}} = -\log \frac{\sum_{\mathbf{x}\in\mathcal{V}} \hat{O}(\mathbf{x})O(\mathbf{x})}{\sum_{\mathbf{x}\in\mathcal{V}}[\hat{O}(\mathbf{x}) + O(\mathbf{x}) - \hat{O}(\mathbf{x})O(\mathbf{x})]}$$

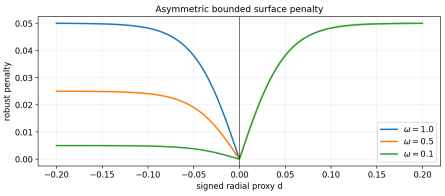


For $e_1 = e_2 = 1$ we obtain
$$\hat{O} = I_1 + I_2 - I_1 I_2.$$

Local surface alignment

$$\mathcal{L}_{\mathrm{SDF}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{p=1}^{P} w_{ip}\, \psi\big(d(\mathbf{x}_i; \Theta_p)\big)$$

- w_{ip} gives the soft assignment of point i to primitive p.
- d is a signed radial **surrogate**, not the Euclidean SDF.
- ψ provides a bounded, asymmetric penalty that limits outlier influence.



$$\psi(d) = \begin{cases} \delta \tanh(d/\delta), & d \geq 0, \\ -\omega\delta \tanh(d/\delta), & d < 0. \end{cases}$$

Compact decompositions require explicit regularization

$$\mathcal{L} = \lambda_{\text{IoU}}\mathcal{L}_{\text{IoU}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_o\mathcal{L}_{\text{overlap}} + \lambda_s\mathcal{L}_{\text{sparsity}} + \lambda_e\mathcal{L}_{\text{exist}}$$

Overlap

$$\mathbb{E}_{\mathbf{x}}\left[\text{ReLU}(\sum_p I_p(\mathbf{x}) - 1)\right]$$

Penalizes excessive overlap between primitives.

Sparsity

$$m_p = \frac{1}{N}\sum_i w_{ip}$$

Discourages unnecessary primitives and favours compact decompositions.

Existence

$\hat{e}_p = \mathbf{1}[m_p > \epsilon_e]$

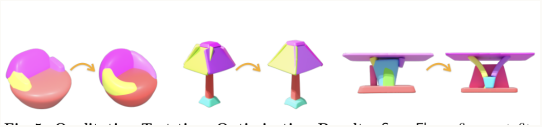
Binary cross-entropy aligns each existence prediction with primitive usage.

Optional test-time optimization

1. Select the primitives whose predicted existence satisfies $e_p > 0.5$.
2. Optimize their geometry Θ_p by Adam, holding the set fixed.
3. Replace pointwise assignments with a global distance to the union.

$$\widetilde{d}(\mathbf{x}) = -T \log \sum_{p \in \mathcal{A}} \exp\left(-\frac{d_p(\mathbf{x})}{T}\right)$$

Smooth approximation to $\min_{p \in \mathcal{A}} d_p(\mathbf{x})$.



Paper Fig. 5. Refines existing primitives rather than training a new network.

IoU rises from 0.72 to 0.87, at a cost of 0.0082 s versus 5 s per object.

Partial scans



Paper Fig. 6. Partial inputs and reconstructions from standard and robust SuperFlex.

1. On complete clouds, predict and optimize high-quality primitive sets.
2. Treat the resulting sets as pseudo-ground truth.
3. Fine-tune on occluded inputs to reconstruct complete primitive structure.

Hungarian matching resolves the arbitrary ordering of the predicted primitives before Chamfer-based supervision is applied to corresponding surfaces.

Experimental protocol

Object-level evaluation on ShapeNet

- 13 object categories.
- 4096 sampled points/object.
- 16 maximum primitive queries.
- 1000 training epochs.
- 4 $\times$ NVIDIA GH200 GPUs reported for training.

Partial observations

- ASE provides partial observations with complete geometry for quantitative evaluation.
- ScanNet++ supplies real-world scenes for qualitative evaluation.
- Metrics include IoU, F-score, Chamfer L1/L2, primitive count and runtime.

Separate feed-forward inference from the much slower optional refinement.

Quantitative comparison for ShapeNet

Method	IoU ↑	F ↑	# primitives ↓	Runtime ↓
Marching Primitives	0.56	0.18	24.10	163.41 s
SuperDec	0.59	0.28	5.79	0.0075 s
SuperFlex (rigid)	**0.70**	0.35	6.04	0.0075 s
SuperFlex (deformable)	**0.72**	**0.37**	**5.64**	0.0082 s

Largest advance

With rigid primitives, the new loss raises IoU from 0.59 to 0.70.

Added expressiveness

Tapering and bending raise IoU from 0.70 to 0.72 while preserving a compact representation.

Paper Table 1, ShapeNet. Forward passes only; all numbers are paper-reported, not reproduced here.

Which loss terms matter?

Supervision	IoU ↑	L_2 ↓	Overlap ↓	# prim. ↓
SuperDec loss	0.59	0.050	4%	5.78
$\mathcal{L}_{\text{IoU}}$ only	**0.74**	0.124	32%	5.94
$\mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{SDF}}$	0.72	**0.042**	14%	5.80
$+ \mathcal{L}_{\text{overlap}}$	0.72	0.043	**8%**	**5.64**

- **IoU alone** greatly improves volumetric reconstruction but produces excessive overlap.
- **The SDF term** improves surface precision, reducing L_2 from 0.124 to 0.042.
- **The overlap term** reduces redundancy from 14% to 8%.

Paper Table 4. Sparsity and existence are held fixed; the IoU score does not improve monotonically with every added term.

Why keep deformation and test-time optimization?

Deformation ablation (refined)

Taper/Bend	IoU	L_2
Neither	0.81	0.041
Taper only	0.83	0.038
Bend only	0.84	0.037
Both	**0.85**	**0.036**

Test-time optimization

	No TTO	With TTO
IoU	0.72	**0.87**
F-score	0.37	**0.49**
Runtime	**0.0082 s**	5 s

Refinement improves geometry, learned feed-forward inference preserves speed.

Occluded objects



Paper Fig. 7: qualitative ScanNet++ scene reconstructions.

ASE partial-cloud method	IoU
SuperDec	0.20
SuperFlex, base	0.20
+ occlusion fine-tuning	0.48
+ ASE fine-tuning	**0.54**

Interpretation

The model completes unseen geometry through priors learned from high-quality full-object decompositions. Better fitting on complete inputs alone does not solve occlusion.

Conclusions

SuperFlex brings geometric abstraction closer to accurate 3D reconstruction.

A complex object can be approximated by a small collection of explicit and interpretable parts. SuperFlex shows that this compact representation can preserve substantially more geometry than previous rigid superquadric decompositions. The main advance does not come from replacing the Transformer backbone. It comes from improving how primitives are fitted. The joint volumetric and surface loss produces more coherent reconstructions, while bending and tapering allow a single primitive to follow curved or asymmetric parts. Beyond complete point clouds, the method extends this representation to partially observed objects. The missing geometry is inferred from learned shape priors rather than measured directly.

Appendix A – Exact z-axis deformation equations

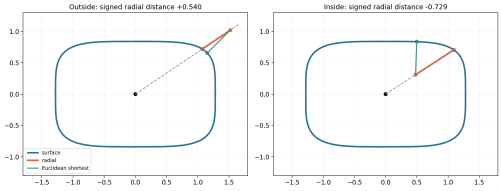
$$T^{-1}(x, y, z) = \begin{pmatrix} \frac{x}{1+\tau_x z/s_z} \\ \frac{y}{1+\tau_y z/s_z} \\ z \end{pmatrix}.$$

$$B_z^{-1}(x, y, z) = \begin{pmatrix} x - (A - r)\cos\alpha_z \\ y - (A - r)\sin\alpha_z \\ \gamma/k_z \end{pmatrix}, \quad \begin{aligned} A &= x\cos\alpha_z + y\sin\alpha_z, \\ \gamma &= \operatorname{atan2}(z, k_z^{-1} - A), \\ r &= k_z^{-1} - \sqrt{z^2 + (k_z^{-1} - A)^2}. \end{aligned}$$

Paper Appendix A. Coordinate branches and $k = 0$ limits need care; atan2 makes the intended quadrant choice explicit.

Appendix B – On the limits of radial distance

- A rigid superellipsoid satisfies $q(\lambda x) = \lambda^{2/\epsilon_1} q(x)$.
- Exact ray intersection at $\lambda_* = q(x)^{-\epsilon_1/2}$; radial displacement $\|x\|(1 - \lambda_*)$.
- Not the Euclidean SDF because nearest surface point need not lie on the center ray.
- With nonlinear tapering/bending, $q(D^{-1}(\lambda x))$ generally is not homogeneous; substitution into the same closed form is a surrogate even for the radial ray intersection.



Paper Eq. (3) describes this as radial distance. Geometric qualification follows from the lack of homogeneity after deformation.

Appendix C – Matching and occlusion loss

$$\mathcal{L}_{\text{geom}} = \sum_{p \in \text{matched active}} \text{CD}\Big(\mathcal{S}(\hat{\Theta}_p), \mathcal{S}(\Theta_p)\Big).$$

$$\boxed{\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{geom}} + \lambda_{\text{IoU}}\mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{rigid}}}$$

- Use a Hungarian one-to-one assignment between the predicted and pseudo-ground-truth primitive sets.
- $\mathcal{L}_{\text{rigid}}$ compares counterparts with deformations switched off.
- Necessary because different parameterizations, symmetries and permutations can produce similar surfaces.

Paper Sec. 4.3 and Eq. (12).

Appendix D – Implementation details

Training choices

Point count N	4096
Primitive queries P	16
Feature width D	128
Decoder layers L	3
Epochs	1000
Initial learning rate	3×10^{-4}
Robust fine-tuning LR	10^{-4}
IoU temperature	10^{-3}

Reproducing authors' inference

```
git clone https://github.com/
GabrielTavernini/superflex

pip install -r requirements.txt
pip install -e .
bash
scripts/download_checkpoints.sh
python demo_viser.py
```

Appendix E – From a single image to 3D primitives



Fig. 13: SuperFlex from a single RGB Image. From an input RGB frame, we first compute instance point maps using SAM3 and MoGE. Then, we apply SuperFlex on top of each object's point map.

- SAM 3 masks instances in the original RGB image.
- MoGE produces partial 3D point maps from a single view.
- Robust SuperFlex turns partial object point sets into complete primitive abstractions.

Paper supplementary Fig. 13. This is an application, not the core inference pipeline or an additional benchmark result.

References and resources

- Tavernini et al., *SuperFlex: Deformable Superquadrics for Point Cloud Decomposition*, ECCV 2026, arXiv:2607.01015.
- Fedele et al., *SuperDec: 3D Scene Decomposition with Superquadric Primitives*, ICCV 2025.
- Barr, *Superquadrics and Angle-Preserving Transformations*, IEEE CG&A, 1981.
- Liu et al., *Marching-Primitives: Shape Abstraction from Signed Distance Function*, CVPR 2023.

- SuperFlex project: `https://superflex3d.github.io/`
- Source and checkpoints: `https://github.com/GabrielTavernini/superflex`

All source-paper figures included for scholarly presentation and credited on their slides. Notebook illustrations are original.